	Year Ended December 31,		
	2019	2018	2017
Cost of revenue	$ 6,403	$ 7,312	$ 5,312
Research and development	44,855	57,188	54,123
Sales and marketing	11,585	14,726	14,959
General and administrative	14,896	17,783	17,187
Total stock-based compensation expense	$ 77,739	$ 97,009	$ 91,581

The weighted-average grant date fair value of stock options granted during 2017 was $2.00 per share. There were no stock options granted in 2019 or 2018. The total grant date fair value of stock options that vested during 2019, 2018, and 2017 was $2.8 million, $10.1 million and $20.2 million, respectively. As of December 31, 2019, the total unrecognized compensation expense related to unvested stock options was $0.3 million, which the Company expects to recognize over an estimated weighted average period of 0.2 years. As of December 31, 2019, the total unrecognized compensation expense related to unvested RSUs was $94.2 million, which the Company expects to recognize over an estimated weighted average period of 1.9 years. As of December 31, 2019, there was zero unrecognized compensation expense related to unvested warrants.

Stock-based compensation is measured at the grant date based on the fair value of the award and is recognized as expense over the requisite service period, which is generally the vesting period of the respective award. The Company accounts for forfeitures as they occur. The fair value of RSUs without market conditions is the fair value of the Company's Class A common stock on the grant date. The fair value of RSUs with market conditions is estimated using a Monte Carlo simulation model. In determining the fair value of the stock options, warrants and the equity awards issued under the 2015 ESPP, the Company used the Black-Scholes option-pricing model and assumptions discussed below. Each of these inputs is subjective and generally requires significant judgment.

Fair Value of Common Stock—The fair value of the shares of common stock underlying stock options had historically been established by the Company's board of directors. Following the completion of the IPO, the Company began using the market closing price for the Company's Class A common stock as reported on the New York Stock Exchange.

Expected Term—The Company does not have sufficient historical exercise data to provide a reasonable basis upon which to estimate expected term due to the limited period of time stock-based awards have been exercisable. As a result, for stock options, the Company used the simplified method to calculate the expected term, which is equal to the average of the stock-based award's weighted average vesting period and its contractual term. The expected term of the 2015 ESPP was based on the contractual term.

Volatility—The Company estimates the expected volatility of the common stock underlying its stock options at the grant date. Prior to 2018, the Company estimated the expected volatility of the common stock underlying stock options, warrants and equity awards issued under its 2015 ESPP at the grant date by taking the average historical volatility of the common stock of a group of comparable publicly traded companies over a period equal to the expected life. The Company used this method because it had limited information on the volatility of its Class A common stock because of its short trading history. Beginning in 2018, the Company used a combination of historical volatility from its Class A common stock along with historical volatility from the group of comparable publicly traded companies.

Risk-Free Rate—The risk-free interest rate is estimated average interest rate based on U.S. Treasury zero-coupon notes with terms consistent with the expected term of the awards.

Dividend Yield—The Company has never declared or paid any cash dividends and does not presently plan to pay cash dividends in the foreseeable future. Consequently, it used an expected dividend yield of zero.

The assumptions used in calculating the fair value of the stock-based awards represent management judgment. As a result, if factors change and different assumptions are used, the stock-based compensation expense could be materially different in the future. The fair value of the stock option awards, warrants, awards issued under the 2015 ESPP, and awards granted to employees was estimated at the date of grant using a Black-Scholes option-pricing model. The fair value of the RSUs with market conditions were estimated using a Black-Scholes option-pricing model combined with a Monte Carlo simulation model. The fair value of these awards were estimated using the following Black-Scholes assumptions: